Project Report
On
Canteen Management System

Submitted by:
Lakshita Kalra (2401010117)
Rahul Kumar (2401010104)
Rahul Bagoria (2401010286)
Vidhi (2401010146)

Department of
Computer Science and Engineering

School of Engineering and Technology
K.R.Mangalam University, Gurugram- 122003

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DECLARATION

We hereby declare that the project work entitled "Canteen Management System" is an authentic record of our work carried out under the guidance of DR.Aarti mam as a requirement of the Web Development Project in the 3rd Semester in the School of Engineering and Technology, K.R. Mangalam University, Gurugram.

The results embodied in this project have not been submitted to any other University or Institute for the award of any degree or diploma.

Lakshita Kalra
Rahul Kumar
Rahul Bagoria

CERTIFICATE

This is to certify that **Lakshita**, **Rahul and Rahul** students of BTECH CSE CORE , 3rd Semester of K.R Mangalam University, Gurugram have completed their Web development project for the Third semester.

Supervisor
Dr.Aarti Mam

ABSTRACT

This project presents the design and development of a modern, responsive **Canteen Management System**, aimed at automating and streamlining daily canteen operations within an educational institution. The system offers a user-friendly, interactive platform that enables students and staff to view menus, place orders, and receive bills digitally, while administrators can efficiently manage food items, pricing, stock levels, and ongoing orders.

Developed using **HTML5, CSS3, and JavaScript** for an intuitive and dynamic frontend, the application incorporates **Node.js and Express.js** on the backend to ensure secure, real-time data processing. A **MongoDB database** is used to handle essential functions such as order records, menu updates, inventory tracking, and user information. Responsive design principles ensure smooth functionality across mobile, tablet, and desktop devices, enhancing accessibility and user convenience.The project was executed through a structured approach that included requirements gathering, interface prototyping, backend routing, and database modeling. Rigorous testing validated system performance, speed, and reliability across various user scenarios, ensuring minimal errors and seamless interaction.

Overall, the Canteen Management System achieves its goal of creating a **fast, efficient, and automated digital solution** that reduces manual effort, minimizes human errors, and improves service delivery. Future enhancements may include features like online payment integration, order analytics, daily sales reports, and role-based dashboards to further strengthen system capabilities. This project showcases how web technologies can transform traditional canteen operations into a smarter, more organized, and user-centric digital platform.

INTRODUCTION

The **Canteen Management System** is a **web-based application** developed to simplify and automate the daily operations of a canteen. In many educational institutions and workplaces, canteen management is still done manually, which often leads to issues like I incorrect order handling, delayed service, and poor record maintenance.
To overcome these challenges, this project aims to create an **efficient, user-friendly, and digital platform** that allows users to view menus and place orders all in one system. The system provides separate interfaces for **customers and staff** ensuring smooth communication and management across all roles.

Developed using **HTML, CSS, JavaScript, Node.js, and Express.js**, this project not only focuses on functionality but also ensures a responsive and intuitive design."The backend, developed using Node.js and Express.js, securely manages data and communicates with the frontend through REST APIs.The database ensures real-time data storage for menu, orders, and users."

Through this project, the team learned the complete process of **web application development**, including frontend design, backend integration, database connectivity, and deployment. It serves as a step toward understanding how real-world systems work and how technology can make

Introduction to the Problem:

In most canteens, operations are still managed manually using paper records and verbal communication. This leads to delays in order processing and errors in billing or order delivery.

It becomes difficult to handle multiple orders during busy hours efficiently.There is no centralized record of sales, orders, or expenses for analysis. Manual systems increase the chances of data loss or mismanagement. To overcome these challenges, an automated web-based Canteen Management System is required to streamline canteen operations.

Solution to the Problem:

- The **Canteen Management System** is a **web-based application** designed to automate the entire process of canteen operations.
- It allows **customers and staff to log in separately**, ensuring role-based access and secure data management.
- The staff can update the menu, manage orders, and monitor inventory in real time.
- The system promotes **faster service, transparency, and**
- **better management**, improving customer satisfaction and business performance.

Objectives:

1.To develop a **full-stack web-based canteen management system** that automates the process of ordering and billing.

2.To implement a backend using **Node.js and Express.js** for dynamic data management and secure communication between frontend and database.

3.To provide **separate login portals** for customers and staff for secure and role-based access.

4.To help **canteen staff manage inventory**, track available food items, and update menus in real time.

5.To **reduce manual errors** in order processing, billing, and inventory management.To improve overall efficiency and customer satisfaction through faster and smoother service.

6.To promote a paperless and eco-friendly management approach.

METHODOLOGY

1. Requirement Analysis
Identified user needs and key features like login, menu display, order, and inventory management.

2. System Design
Planned the system structure, defined database tables and created flowcharts and wireframes for easy usability.

3. Development
Built the frontend using HTML, CSS, and JavaScript, and backend using Node.js and Express.js. Connected the frontend with the backend for real-time data exchange.

4. Implementation & Testing
Verified API calls, ensured data is correctly stored/retrieved from the backend database, and tested all modules for smooth client-server communication.

5. Deployment
Frontend deployed on Netlify and backend hosted locally or on

WORKFLOW

1. **User Login / Registration**
   Users (customers and staff) register or log in using their credentials to access the system.

2. **Home / Dashboard**
   After login, users are redirected to their respective dashboards — customers see the menu, and staff manage orders and inventory.

3. **Menu Management**
   Staff can add, update, or remove food items and set prices. Customers can view the available items.

4. **Order Placement**
   Customers select food items and place orders through the website interface.

5. **Order Processing**
   The system sends order details to the staff for preparation and updates the order status in real time.

6. **Billing & Payment**
   The total bill is generated automatically, reducing manual calculation errors. Customers can view or print their receipts.

7. **Logout / Exit**
   Users can securely log out from their accounts after completing transactions.

TOOLS AND TECHNOLOGIES

1. Frontend Technologies

- HTML, CSS, JavaScript — for structure, design, and interactivity.

2. Backend Technologies

- **Node.js** – runtime environment for executing server-side JavaScript.
- **Express.js** – lightweight framework to build RESTful APIs.

3. Database

- **JSON file / MongoDB** – used to store user, menu, and order data persistently.

4. Development Tools

- VS Code
- Git & GitHub
- Chrome (testing)

Hosting Platforms

- Netlify (frontend)
- Render(backend)

These technologies were chosen for their **simplicity, reliability, and compatibility** across platforms.

The combination of HTML, CSS, and JavaScript allows for a **responsive and user-friendly interface**, while LocalStorage ensures **quick data access without requiring a server connection**.

Overall, the selected tools make the system **lightweight, easy to deploy, and efficient for small-scale canteen operations**.

APPLICATIONS

1. Can be used in **college and office canteens** to manage orders, billing, and inventory efficiently.

2. Useful for **small food outlets** or **cafeterias** to reduce manual record-keeping.

3. Helps canteen staff track menu items, stock levels, and customer orders in real time.

4. Customers can **view menus and place orders** easily through a digital interface.

5. Can be adapted for school, hospital, and hostel canteens with minimal customization.

6. Provides a base for **future integration** with online payment systems and live order tracking.

BENEFITS

1. **Time-Saving:** Automates order management, reducing waiting time for customers.

2. **Error Reduction:** Minimizes manual mistakes in billing and record-keeping.

3. **Easy Management:** Simplifies tracking of orders, menu items, and inventory.

4. **User-Friendly Interface:** Provides a simple and attractive layout for both customers and staff.

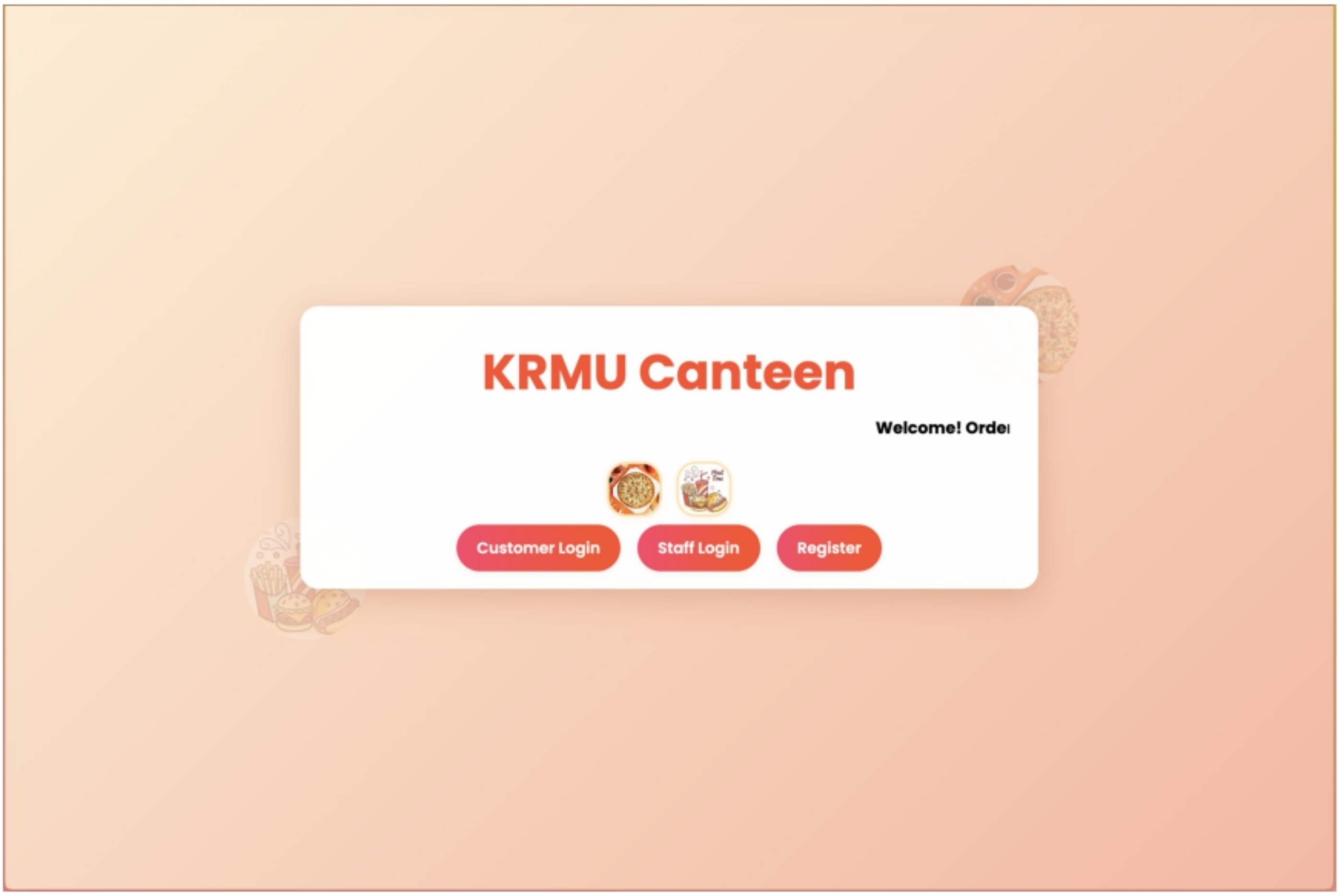
5. **Paperless Operation:** Promotes an eco-friendly and efficient digital process.

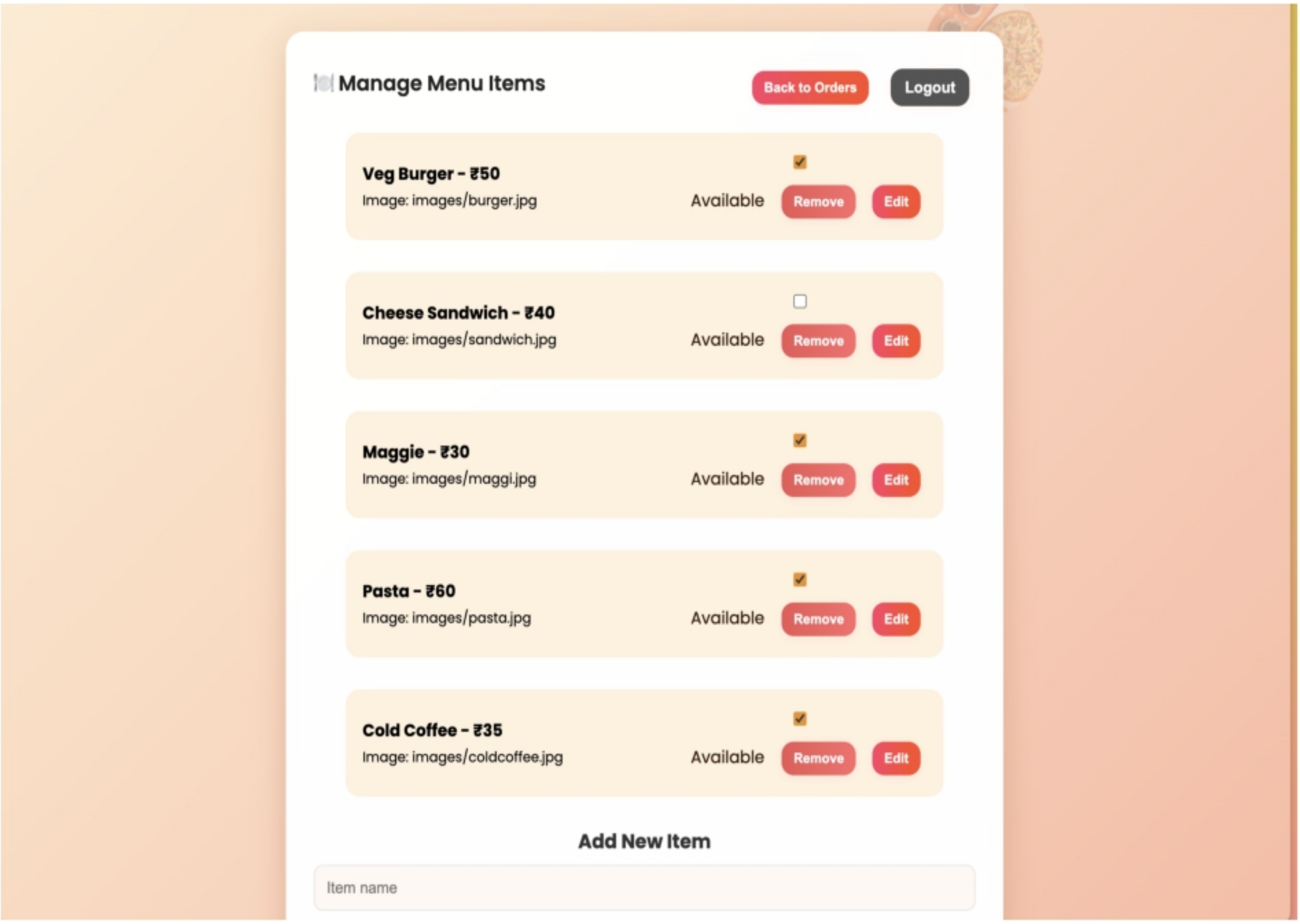
6. **Better Customer Experience:** Ensures faster service and improved satisfaction.

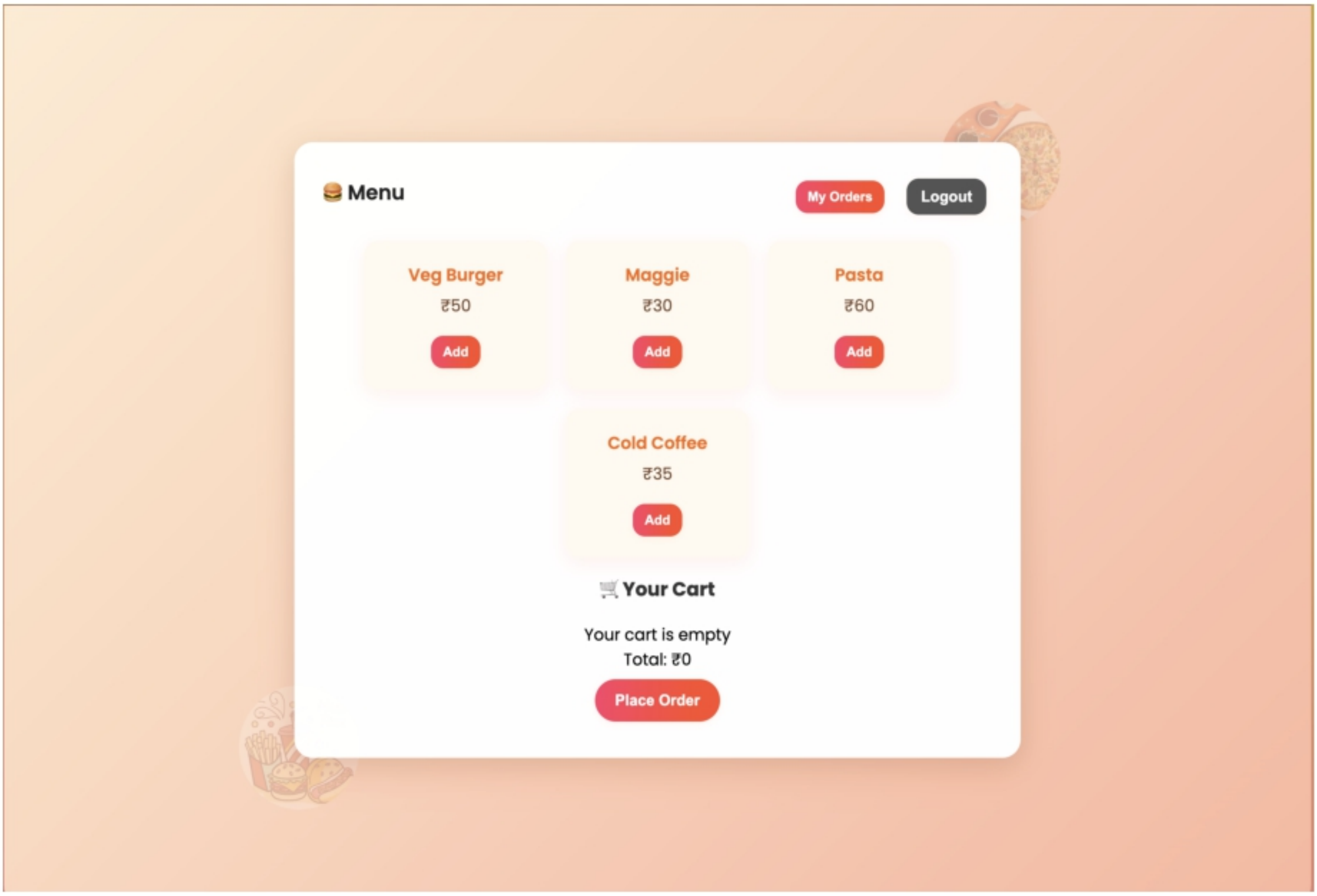
7. **Scalability:** Can be easily extended to support online payments and live order tracking in the future.

8. **Real-Time Functionality:** Backend integration ensures that all orders and menu updates are stored and fetched instantly.

PROJECT SNAPSHOTS




Manage Menu Items
Back to Orders
Logout
Veg Burger - ₹50
Image: images/burger.jpg
Available
Remove
Edit
Cheese Sandwich - ₹40
Image: images/sandwich.jpg
Available
Remove
Edit
Maggie - ₹30
Image: images/maggi.jpg
Available
Remove
Edit
Pasta - ₹60
Image: images/pasta.jpg
Available
Remove
Edit
Cold Coffee - ₹35
Image: images/coldcoffee.jpg
Available
Remove
Edit
Add New Item
Item name


Menu
My Orders
Logout
Veg Burger
₹50
Add
Maggie
₹30
Add
Pasta
₹60
Add
Cold Coffee
₹35
Add
Your Cart
Your cart is empty
Total: ₹0
Place Order

CHALLENGES

1. **API Testing:** Ensuring seamless data flow between frontend and backend.

2. **Database Connection:** Managing persistent storage and handling asynchronous data operations.

3. **Deployment:** Hosting both frontend and backend on separate platforms.

4. **CORS and Routing Issues:** Handling communication between different servers securely.

CONCLUSION

The **Canteen Management System** successfully automates the basic operations of a canteen, such as order placement, billing, and menu management. It provides an **easy-to-use, efficient, and paperless solution** that saves time for both customers and staff. By using web technologies like HTML, CSS, and JavaScript, the system ensures a smooth and responsive experience. Although it currently uses LocalStorage for simplicity, it lays a strong foundation for future enhancement with real-time databases and online payment integration.

REFERENCES

- **MySQL Documentation** – MySQL Developer Guide
  https://dev.mysql.com/doc/
- **W3Schools** – Tutorials on HTML, CSS, JavaScript, and PHP
  https://www.w3schools.com/
- **GeeksforGeeks** – Programming and Database Concepts
  https://www.geeksforgeeks.org/
- **Stack Overflow** – Community Discussion and Code Debugging
  https://stackoverflow.com/
- **Research Paper:**"Automation of College Canteen Management System"–International Journal of Innovative Research in Computer and Communication Engineering , Vol. 8, Issue 4, April 2020.

GITHUB LINK:

https://github.com/lakshita-kalra/Canteen-Webdev